Rudra Pratap Singh

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Education

Vellore Institute of Technology — Tamil Nadu, India
Bachelor of Technology in Computer Science and Engineering — *July 2026 – May 2030*

Experience

In-Charge, Mathematics Club | *School Leadership*

- Managed a team to design and run an interactive mathematics exhibition, taking responsibility for planning and coordination
- Communicated complex concepts to a general audience through hands-on exhibits

Academic Competitions | *Science Olympiad, Quizzing*

- Won a medal in a national-level Science Olympiad
- Competed as a regional-level quiz champion, solving problems quickly under strict time pressure

Projects

AirChirp | *HTML5, JavaScript, Web Audio API, pako.js* — August 2026

- Developed an air-gapped acoustic modem running in the browser to transfer files via near-ultrasonic sound waves (17.5kHz–19.5kHz)
- Implemented multi-carrier modulation across four parallel frequency lanes to stream data using laptop speakers and microphones
- Secured data payloads using client-side AES-256-GCM encryption and compressed data streams with pako.js
- Visualized real-time Fast Fourier Transform (FFT) frequencies using the HTML5 Canvas API and Web Audio AnalyserNode

TabCluster | *JavaScript, WebRTC, PeerJS, Web Workers, HTML5 Canvas, Git* — Sep 2026

- Built a browser-based distributed compute engine in a 48-hour hackathon with a team of 3, letting phones and laptops join a zero-server P2P cluster by scanning a QR code or entering a room code
- Offloaded Mandelbrot tile computation to Web Worker threads and returned results as binary `Uint8ClampedArray` buffers, keeping the UI responsive and avoiding serialization overhead
- Implemented a pull-based task queue so faster devices process more tiles, with timeout-based re-queueing of chunks from disconnected workers
- Developed a live host dashboard showing active workers, completion rate, and estimated TFLOPS, painting tiles onto an HTML5 Canvas as they arrive

Courses

CS50x: Introduction to Computer Science – Harvard University (HarvardX)
CS50P: Introduction to Programming with Python – Harvard University (HarvardX)
CS50W: Web Programming with Python and JavaScript – Harvard University (HarvardX)

Skills

Languages: Python, C, SQL, JavaScript, HTML/CSS
Frameworks: React, Node.js, Flask
Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm
Libraries: pandas, NumPy, Matplotlib, pako.js, WebRTC
Soft Skills: Team Leadership, Adaptability, Problem Solving, Quick Learning

Interests

Video Games, Sports, Traveling